

RESEARCH STATEMENT

My **research interests** lie in empirical microeconomics where I use modern econometric methodology to answer open economic questions in the fields of labor economics, industrial organization, finance and development economics through using new data sources, tools/methods and approaches. My research is usually motivated by identifying broad trends in markets or industries, understanding how existing systems work, and suggesting process/product improvements. I have a background in causal analysis, applied micro-econometrics, and implementation of randomized control trials. With my micro econometric analysis, I help contribute to the academic literature, optimize business decision-making and formulate public policy recommendations.

My **job market paper** examines the impact of occupational licensing on worker mobility in the US from 2016 to 2020. Using a state, occupation, and time fixed effects model along with detailed level socio-economic controls, I find that occupational licensing does not reduce individual geographic mobility either between states or within a state. In particular, I am passionate about using methods outside the traditional Economics toolkit to answer my research questions and use k-means unsupervised machine learning algorithm to correct individual misreporting in Current Population Survey (CPS) responses and to create a cleaner measure of individual licensing estimate. I have presented this paper at research conferences and expect to submit to a journal shortly, and would like to start a new project on occupational licensing - extending some prior work on impact of stricter mortgage broker regulation on consumer outcomes.

Within labor economics, I have two **ongoing projects** on minimum wages that I got associated with while working as a research assistant to Dr. Aleksandar Sasha Tomic at Boston College during my graduate schooling. First, we have looked at Economic Freedom Index rankings constructed by Dean Stansel (2019) and improve upon his measurement of labor market freedom component through addressing minimum wage bindingness issue across different labor markets. Second, I created a MSA level panel dataset using Occupational Employment and Wage Statistics (OEWS) data from Bureau of Labor Statistics (BLS) and Personal Income data from Bureau of Economic Analysis (BEA) to study impact of minimum wages on inter-generational social mobility using Raj Chetty's quintile transition matrices of child and parent income. While we have already presented the former work, the second work is still ongoing, and I will be finishing these projects soon.

In my **second dissertation chapter**, I collaborate with Dr. Nehan Naim (Assistant Professor at SUNY-ESF) who was my former colleague at Northeastern University to study the pricing strategy of electric vehicle charging station (EVCS) in 2019. Using a logit specification, we examine whether a station is more likely to charge a price based on competition (number of other stations in the vicinity), station level characteristics (technology and number of charges), business partnerships among electric vehicle chains (roaming/non-roaming networks), and market demographics after controlling for chain fixed effects. As in industrial organization literature, we find evidence for stations exploiting their market power and charging a positive price when they have a high-quality product (L3 fast charging technology), are less capacity constrained (more chargers), are chain affiliated (not an independent) and have formed a business partnership to become part of the roaming network (network characteristic). We have submitted this to paper to a Contemporary Economic Policy journal.

In the future, I plan to **extend** the paper in a few different directions due to our rich data. First, I will transform the current static competition model into a dynamic structural model where entry/exit decisions of firms can also be accounted for like Mazzeo (2002) does for entry of high/low quality motel across highways exits to study product and price differentiation. Second, like Seim's (2006) modelling of video rental retail industry, I will focus on spatial differentiation by stations to see if firms use location choices to shield themselves from competition. Third, once I collect some electric vehicle car sales data and combine that with our station level data, we can conduct some welfare studies about the impact of electric vehicle adoption in US.

In my **third dissertation chapter**, I look at impact of FinTech firms on US housing mortgage markets from 2011 to 2017. While FinTech firms could be causing more refinancing, they could also be simply filling in the refinancing need better through enhanced prediction of potential customers and aggressive expansion in geographic areas wanting refinancing, say through online advertising. Building upon the work of Fuster et al (2016), I better address this endogeneity issue with an instrumental variables approach (state-level Batik instrument for the number of approved applications by FinTech firms in a county). Using a year-county fixed effects model at county level, I find significantly higher number of approved home refinancing and home purchases applications in counties with a higher FinTech market share, even after controlling for aggregated loan, borrower, and county level controls, presumably due to FinTech "originate to distribute" model. I would

like to extend this work which currently is limited to the largest 20 firms in 2016 to include all firms, and thus distinguish between business stealing and market expansion effects.

In the future, I am interested in continuing to study topics in finance and housing market. First, as a trading enthusiast, I would like to conduct empirical finance research on returns to trading strategies and back test their performances in Python. Second, I would like to empirically analyze the factors that cause assessing jurisdictions to have regressivity in their property tax system within Massachusetts, a project I had briefly worked on with Dr. Gustavo Vicentini. Regressivity, a specific pattern of vertical inequity, is defined as when lower valued properties are over-assessed compared to higher valued properties, and can be operationalized using sale transaction data on single family residences.

Currently, my **pet project** is to estimate demand by running a BLP (Barry-Levinson-Pakins) model on the telecom sector in India. I have set up a team consisting of my graduate students at Boston College that is scrapping online price data of telecom firms in different markets using Python, and also cleaning wireless services quality and quantity data. I pulled these students from my classes into the research projects - mentoring and training them to conduct Econometric analysis in software tools including R, Stata and/or Tableau. Further, I also supervise students in my Econometrics classes on their term projects, and lead a Directed Practicum course.

I have also worked as a **research assistant** in small and large sized teams. For example, I worked with Dr. Catalina Herrera Almanza at Northeastern University and her co-PIs at Boston University (Dr. Mahesh Karra) and Boston College (Dr. S Anukriti) to conduct summary statistics of Jaunpur Social Networks Study (JSNS) – a randomized control experiment that explores the causal impact of a social network-based family planning intervention on women's contraceptive use, reproductive health, fertility, and well-being in rural India. Additionally, I was part of Seminar Dynamics Collective, a group of 97 members of the economics community who collected and processed data on interaction between presenters and their audience in research seminars to test for gender neutrality in economic seminars.

Some topics that I have also examined include looking at anti-trust literature for merger analysis review, conducting merger simulations in R and helping create airline industry trends using T-100 Traffic Reports and DB1B Ticket-Level Domestic database for Dr. John Kwoka. Additionally, I have created health datasets to study impact of obesity on C-section operations in Mexico for Dr. Silvia Prina.

I am also interested in **business research**. At Shell TechWorks as a commercialization and deployment intern, I was involved with net present value calculations and financial modelling of multiple business projects, cost estimation of well abandonments, generating market feasibility report about investing in nascent technologies and deal validation/compliance. Also, I have worked as a business-cum-operations analyst at small to medium sized enterprises in Hong Kong, India and USA to increase business profitability in sectors ranging from medical clinics to tourism companies. I enjoy documentation of systems and processes, benchmarking performance, and even have some bank stress testing work experience.

In summary, my research experience lies at the intersection of microeconomics and empirical econometrics, and I have studied topics in industrial organization, labor economics, finance and development economics. With my background in causal analysis and applied micro-econometrics, I like to work in teams and collaborate on my projects. I have experience in microeconomic modelling, working with large real-world data sets to building scalable models, helping businesses optimize to increase their profitability, and helping create better public policy.